

Azure Databricks Unity Catalog

1. Data Governance

Explanation:

A framework of rules, policies, and controls to manage data securely and consistently across an organization.

Example:

Only the HR team can view employee salary data. Marketing and interns cannot.

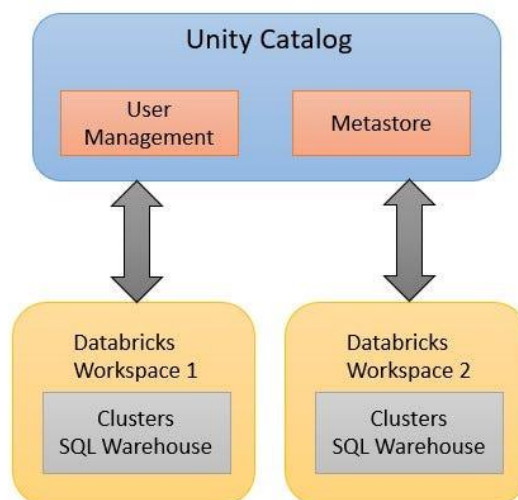
2. Unity Catalog

Explanation:

Databricks' centralized governance solution that manages data access, permissions, metadata, and audit logs across all workspaces.

Example:

A single Unity Catalog controls who can see customer data across Workspace A, B, and C, instead of managing each workspace separately.



3. Metastore

Explanation:

A central repository that stores metadata (data about data) like table names, owners, schemas, permissions, and storage locations.

Example:

A librarian's register that tracks book names, authors, and who borrowed them. The register is the metastore; the books are the actual data.

4. Workspace

Explanation:

Your Databricks environment where you write notebooks, run clusters, and manage projects.

Example:

A personal desk in an office where you keep your laptop, notebooks, and tools to do your daily work.

5. Namespace

Explanation:

An organized naming structure to group and locate data assets easily.

Example:

Your laptop folders:

Documents → Work → Reports

This organization helps you find files quickly.

6. Three-Level Namespace (Catalog → Schema → Table)

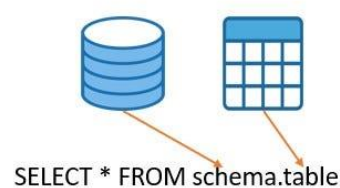
Explanation:

A hierarchical structure to uniquely identify every data asset.

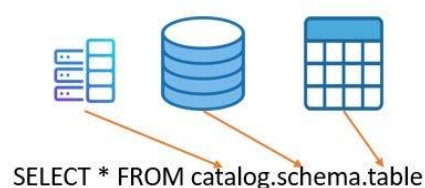
Level Object Analogy

- 1 **Catalog** Country / Company Building
- 2 **Schema** State / Room inside the building
- 3 **Table** City / Cabinet inside the room

Traditional SQL Two-Level Namespace



Unity Catalog Three-Level Namespace



Example:

Sales.Customers.customer_master

- Sales = Catalog (department building)
 - Customers = Schema (room for customer data)
 - customer_master = Table (cabinet with customer details)
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7. Unity Catalog Objects

Explanation:

Resources managed by Unity Catalog, such as Catalogs, Schemas, Tables, Views, Volumes, Functions, and Models.

Example:

Inside Unity Catalog, you create:

- A Catalog called Finance
 - A Schema called Accounts
 - A Table called invoices
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8. Complete Flow to Set Up Unity Catalog

Azure Storage

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Create Metastore

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Enable Unity Catalog

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Attach Workspace

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Create Catalog → Create Schema → Create Table

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Assign Permissions (e.g., GRANT SELECT TO group)

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9. Quick Memory Trick

- **Governance:** Only Sales team can modify Sales data.
 - **Unity Catalog:** Manages all permissions centrally.
 - **Metastore:** Stores metadata for all catalogs, schemas, and tables.
 - **Workspace:** Data engineers access this structure via notebooks.
 - **Namespace:** Every table has a unique three-level address.
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10. Real-World Example: E-Commerce Company

Unity Catalog

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|— Sales Catalog

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|— Customers Schema → customer_master, customer_orders

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|— Products Schema → product_details, product_inventory

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|— HR Catalog

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|— Employees Schema → employee_details, employee_salary

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|— Finance Catalog

|— Accounts Schema → invoices, payments
